DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
Supplementary Summer Examination – 2025			
Course: B. Tech	**Branch : Chemical**		**Semester : II**
Subject Code & Name: BTES205E - Energy and Environment Engineering			
Max Marks: 60	**Date:10/07/25**		**Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		**12**
1	Which of the following is a base load power plant? a) Gas turbine b) Diesel engine c) Hydro plant d) Nuclear plant	**CO1**	**1**
2	Which energy source is non-renewable? a) Solar b) Biomass c) Wind d) Coal	**CO1**	**1**
3	The main pollutant from thermal power plants is: a) CO_2 b) NH_3 c) CH_4 d) O_3	**CO2**	**1**
4	Which power generation method uses the temperature difference in ocean water? a) Tidal b) Wave c) OTEC d) Geothermal	**CO2**	**1**
5	Which of the following is used in MHD generation? a) Mercury b) Argon c) Ionized gas d) Steam	**CO1**	**1**
6	A thermionic generator converts: a) Heat to light b) Heat to electricity c) Electricity to heat d) Light to electricity	**CO2**	**1**
7	Fuel cells generate electricity through: a) Combustion b) Nuclear reaction c) Chemical reaction d) Magnetism	**CO3**	**1**
8	Which of the following causes soil pollution? a) Wind turbines b) Industrial effluents c) Tidal energy d) None of the above	**CO3**	**1**
9	Which of these is an advantage of nuclear power? a) No radioactive waste b) High carbon emissions c) Low	**CO4**	**1**

	operating cost d) High water consumption		
10	The major cause of water pollution is:	**CO4**	**1**
	a) Noise b) Industrial discharge c) Solar panels d) Biomass		
11	Which component is essential in energy conservation in HVAC systems?	**CO5**	**1**
	a) Chiller b) Thermostat c) Battery d) Switch		
12	Biomedical waste is best disposed of by:	**CO5**	**1**
	a) Landfilling b) Open burning c) Incineration d) Composting		
Q. 2	**Solve the following.**		**12**
A)	Explain the schematic layout of a nuclear power plant and discuss its advantages and disadvantages.	**CO1**	**6**
B)	Describe thermoelectric and thermionic generators with their working principles.	**CO1**	**6**
Q.3	**Solve the following.**		**12**
A)	Describe the working of a solar power generation system. Mention its advantages and disadvantages.	**CO2**	**6**
B)	With a neat diagram, explain the principle of Magneto Hydro Dynamic (MHD) generation.	**CO2**	**6**
Q. 4	**Solve Any Two of the following.**		**12**
A)	Describe energy conservation techniques in electric furnaces and ovens.	**CO3**	**6**
B)	Explain the principle of maximum energy efficiency and cost effectiveness.	**CO3**	**6**
C)	Discuss the methods of energy conservation in fans and blowers.	**CO3**	**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	What are the sources and effects of air pollution on human health?	**CO1**	**6**
B)	Discuss particulate emissions and air quality standards.	**CO2**	**6**
C)	Explain the different methods of controlling air pollution.	**CO3**	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	What are the major causes and effects of water pollution? Suggest	**CO3**	**6**

	suitable control measures.		
B)	Discuss noise pollution: sources, effects, and methods of control.	**CO4**	**6**
C)	Explain thermal pollution and its effects on aquatic life.	**CO5**	**6**
	***** End *****		